

Model and test verification for concrete fatigue failure of the steel-concrete hybrid wind turbine tower

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ABSTRACT

With the widespread application of steel-concrete hybrid wind turbine towers, concrete fatigue design has become a challenge in engineering design and a primary factor affecting costs. Concrete fatigue is influenced by many interrelated nonlinear factors, and there is no consensus on the quantitative measurement of these. To develop more accurate methods for describing the S-N curves of concrete materials and numerical calculation approaches, this study employs a back-propagation neural network (BPNN) to predict concrete fatigue behavior using an expanded experimental database of 558 constant-amplitude fatigue test results. On the test set, the BPNN predictions achieve a coefficient of determination (R^2) of 0.926, a root mean square error (RMSE) of 0.352, and a mean absolute error (MAE) of 0.244, significantly outperforming existing code-based methods (NEN 6723: $R^2 = 0.411$, RMSE=1.894, MAE = 1.640; EN 1992: $R^2 = 0.410$, RMSE=1.312, MAE = 1.085; fib Model Code 2010: $R^2 = 0.056$, RMSE=5.755, MAE = 2.011). Additionally, a 1:5 scale specimen based on an engineering prototype was designed and tested under variable-amplitude high-cycle fatigue loading. Three mean stress correction methods (Goodman, Gerber, and R-ratio) are evaluated in fe-safe to determine the most suitable approach for concrete fatigue analysis. The observed fatigue damage evolution follows a three-stage process—stress redistribution, steady damage accumulation, and rapid failure—which the BPNN-based finite element model accurately captures, achieving an average stiffness error of 9.43% compared to experimental measurements. Therefore, the BPNN method proposed can be applied to the calculation of concrete S-N curves when there is sufficient data support, and the fatigue performance of complexly stressed structures can be effectively designed using fe-safe.

Summary:

- The S-N curves of concrete fatigue were described by using BPNN.
- The best method for handling means stress in concrete fatigue was chosen.
- The fatigue test on the tower based on the engineering prototype was completed.
- The S-N curves of BPNN and codes have been evaluated in fe-safe.

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1. Introduction

With the widespread application of steel-concrete hybrid tower structures in wind energy engineering, the issue of concrete fatigue failure has become a pressing design challenge. The fatigue performance of concrete is particularly critical in wind energy engineering, as turbine towers are subjected to prolonged cyclic wind loading [1–3], which can gradually induce fatigue damage in concrete components, potentially leading to serious safety hazards. Therefore, understanding and predicting concrete fatigue performance under different wind loading conditions is paramount. Unlike other materials such as steel, concrete fatigue failure exhibits strong variability due to its nature as a complex multiphase composite material. Due to the nonlinear mapping relationship between fatigue life and its influencing factors, this makes it challenging to accurately account for its characteristics in design. To address this issue, researchers and engineers are continuously striving to develop more accurate models and methods to assess the fatigue performance of concrete structures under long-term wind loading [4–9]. They employ a combination of experimental testing and numerical simulation to delve into the fatigue behavior of concrete under various loading histories, proposing corresponding design recommendations and improvement measures [10–13].

The main difficulties in determining the fatigue life of concrete are primarily reflected in the complexity of influencing factors and the material's strong nonlinearity. The fatigue strength of concrete is primarily influenced by several parameters. The first parameter is the maximum and minimum stress levels ($S_{\min}$ and $S_{\max}$) during the application cycles. An increase in the minimum stress $S_{\min}$ leads to a higher number of cycles to failure. The second parameter is the strength grade of the concrete. During fatigue testing, a significant scatter in number of cycles to failure (N) at a given stress level is observed, and this scatter is attributed to the variability in the compressive strength of the concrete. While variables such as cement ratio, cement content, entrapped air content, curing conditions, and loading age exert non-negligible effects on concrete fatigue strength, incorporating these factors from an engineering application perspective would introduce excessive model complexity [14–16]. This simplification aligns with current design standards that exclusively account for concrete strength and stress effects.

Traditional fatigue life analysis relies on mathematical statistics, fitting experimental data to a specific regression equation. Initially, the parameters considered in the fatigue life equation only included the stress level. Subsequently, parameters such as the stress ratio (R) [17,18], probability of failure (P) [19], and loading frequency [18,20] have gradually been integrated into practical applications. For example, standards like the former Dutch national code NEN 6723:2009 [21], NEN-EN 1992-1-1+C2:2011 [22], and the fib Model Code 2010 [23] provide S-N curves for concrete. Although the proposed equations are comprehensive and complex, they cannot be applied to all fatigue analyses, and it is difficult to ensure their accuracy. Furthermore, there has not yet been a consensus on the quantitative measurement of these factors. Table 1 provides an overview of code equations utilized in this study, both current and previously employed.

Artificial neural networks (ANNs), with their strong learning and adaptation capabilities, can process large-scale complex data, offering high adaptability, robustness, and excellent nonlinear modeling ability. Compared to formula fitting, ANN can better uncover deep-level information within the data [24,25]. Nowadays, many scholars have applied techniques from the field of deep learning to fatigue analysis of concrete in civil engineering [26–31]. Lantsoght [32] derived two new formulas based on a high-strength concrete

Table 1
An overview of code equations.

Code	Equations
NEN 6723:2009	$\log N = \frac{10}{\sqrt{1-R}} \left(1 - \frac{\sigma_{\max}}{f_{b,v}} \right) \text{ for } \frac{\sigma_{\max}}{f_{b,v}} > 0.25 \quad (1)$
	$R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{S_{\min}}{S_{\max}} \quad (2)$
	$f_{b,v} = \frac{f_{b,rep,v}}{\gamma_m} \text{ with } \gamma_m = 1.2 \quad (3)$
	$f_{b,rep,v} = 0.5f_{b,rep,k} + 12.75 \quad (4)$
NEN-EN 1992-1-1 + C2:2011	$\log N = \frac{14}{\sqrt{1-R}} (1 - E_{cd, \max}) \quad (5)$
	$E_{cd, \max} = \frac{\sigma_{\max}}{f_{cd, \text{fat}}} \quad (6)$
	$f_{cd, \text{fat}} = 0.85\beta_{cc}(t_0)f_{ck} \left(1 - \frac{f_{ck}}{250} \right) \quad (7)$
	$\beta_{cc}(t_0) = \exp \left\{ s \left[1 - \left(\frac{28}{t_0} \right)^{0.5} \right] \right\} \quad (8)$
fib Model Code 2010	$\log N = \begin{cases} \frac{8}{\gamma-1} (S_{\max} - 1) \text{ for } \log N \leq 8 \\ 8 + \frac{8 \ln(10)}{\gamma-1} (\gamma - S_{\min}) \log \left(\frac{S_{\max} - S_{\min}}{\gamma - S_{\min}} \right) \text{ for } \log N > 8 \end{cases} \quad (9)$
	$\gamma = \frac{0.45 + 1.8S_{\min}}{1 + 1.8S_{\min} - 0.3S_{\min}^2} \quad (10)$
	$S_{\max} = \frac{ \sigma_{\max} }{f_{cd, \text{fat}}}, S_{\min} = \frac{ \sigma_{\min} }{f_{cd, \text{fat}}} \quad (11)$
	$f_{cd, \text{fat}} = 0.85\beta_{cc}(t_0)f_{ck} \left(1 - \frac{f_{ck}}{400} \right) \quad (12)$

experimental database: one for assessing the fatigue strength of existing structures and another for simplified design. Chen [33] studied concrete tensile fatigue behavior using a Bayesian regularized backpropagation neural network (BR-BPNN) to predict fatigue life reliably. The BR-BPNN model demonstrated high reliability in predicting fatigue behavior, aiding in fatigue test program design. Abambres [34] utilized artificial neural networks to analyze concrete compression fatigue behavior with 203 datapoints, resulting in an optimal model with a maximum relative error of 5.1% and mean relative error of 1.2%, outperforming existing code expressions and offering improved accuracy for concrete structure design and assessment. In addition, Abambres [35] proposes an artificial neural network (ANN)-based formula to estimate the shear capacity of one-way reinforced concrete slabs under concentrated loads, using a comprehensive database of 287 test results and achieving zero relative errors, which outperforms Eurocode 2 provisions and improves current assessment procedures for reinforced concrete slab bridges. Zhang [36] employs machine learning techniques to address concrete fatigue, exploring algorithms like random forest, support vector machine, and artificial neural networks. By training on extensive experimental data, a model accurately estimating the maximum cycles to failure was chosen, leading to a proposed strength degradation model for evaluating concrete's residual strength post-fatigue damage, crucial for determining structure service life. Xiao [37] demonstrates that ANN are more effective than traditional statistical methods in predicting the fatigue life of rubberized asphalt concrete mixtures containing reclaimed asphalt pavement under various conditions. Vadood [38] evaluated the fatigue life of polypropylene and polyester fiber-reinforced HMA using regression and optimized ANN models, finding that a two-hidden-layer ANN optimized with a genetic algorithm can predict the fatigue life with high accuracy. Amar [39] utilizes an ANN with 18 input parameters to predict the compressive strength of waste-based concretes containing various mineral additions, achieving high accuracy based on a dataset of 1303 different mixtures from 22 bibliographic sources. Abuodeh [40] employs Sequential Feature Selection and Neural Interpretation Diagram to identify critical material constituents affecting the compressive strength of ultra-high performance concrete and develops an ANN model with improved accuracy using four key constituents: cement, fly ash, silica fume, and water. Song [41] develops a fatigue performance prediction model for reinforced concrete beams using a deep belief network optimized by particle swarm optimization, demonstrating its superior accuracy and efficiency compared to single deep belief network and backpropagation models through various evaluation metrics. Liang [42] employs gradient boosting regression tree (GBRT) and multiple linear regression (MLR) models to predict the fatigue life of concrete under repeated compression, demonstrating that the GBRT model, when optimized with classical formulas and feature selection, achieves superior accuracy and reliability compared to traditional methods and European code predictions, offering valuable insights for concrete technical design specifications.

Although there have been numerous research outcomes on the fitting and prediction of fatigue life of concrete materials, there has not yet been a sufficiently detailed analysis and argumentation on how to apply these results in engineering design, and how to integrate the existing research findings with general finite element analysis. In this work, this study combines the existing fatigue experiment database with a powerful neural network to attempt to describe the fatigue life of concrete specimens under compression. In addition, this study uses BPNN to obtain the fatigue life of concrete based on the existing data, and study whether there are significant differences in fatigue performance under the influence of different factors. According to the analysis results, a fatigue constitutive model for concrete is established, and a scaled-down test model designed according to actual engineering specifications is used for validation in fe-safe.

2. Database and methods

2.1. Database

The database used in this paper consists of experimental results from existing published literature [32, 43–50], covering common types of concrete such as normal concrete, high-strength concrete, and steel fiber ultra-high performance concrete (UHPC). Only results from constant-amplitude fatigue tests were included in this study. The maximum compressive strength of the concrete in the database is 226 MPa, and the minimum is 24.0 MPa, as shown in Fig. 1 (a). The data distribution in the database is primarily based on the compressive strength of normal concrete mixture below 100 MPa, while the compressive strength of UHPC concrete is mainly in the range of 160–180 MPa. The lower limit of stress cyclic values in the database ranges from 0 to 0.836, the upper limit ranges from

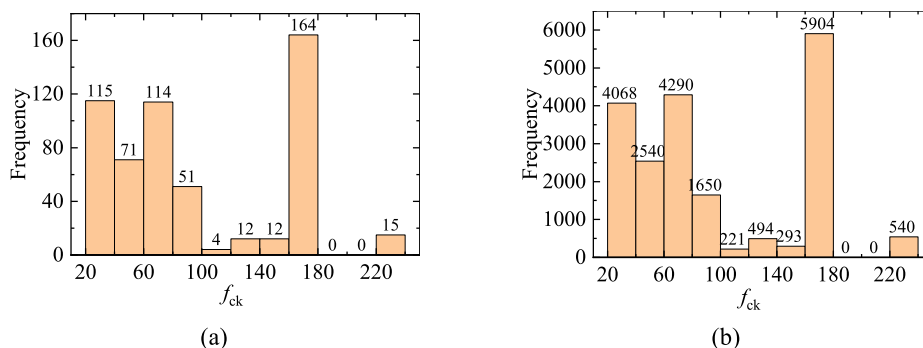


Fig. 1. The distribution charts of concrete compressive strength. (a) the original data, and (b) the expanded data.

0.435 to 0.960, and the number of cycles ranges from 3 to 63,841,047, encompassing both high-cycle and low-cycle fatigue loading conditions.

Additionally, this study expanded the dataset to 20,000 data points by adding white noise to the original data to ensure that the model has good generalization and Effective Regularization [51]. Gaussian white noise with zero mean and a standard deviation of 5% of each parameter's original value was added to the dataset. This noise level was selected to be sufficiently large to enhance model generalization and provide effective regularization, yet small enough to ensure that the augmented data points remain within physically realistic ranges. The distribution of the expanded data is shown in Fig. 1(b). From the Fig. 1(b), the distribution of the expanded dataset is consistent with the original dataset, which provides more possibilities for the subsequent division of the test set and training set. All data in the dataset were mapped to the range [0, 1] using the Min-Max Normalization [52] method to improve the model's performance and convergence speed. It is also worth noting that this study performed a logarithmic transformation on the number of cycles before normalization to reduce the impact of data outliers. The relevant parameters of the original data and the expanded data are shown in Table 2.

The correlation coefficient heat map for the features is shown in Fig. 2. From the Fig. 2, it can be observed that for the concrete fatigue life ($\text{Log}_{10}(N)$), both f_{ck} and S_{min} show a positive correlation, indicating that an increase in the concrete strength grade has a beneficial effect on fatigue life. But the positive correlation value of S_{min} is closer to 0 compared to the other parameters, indicating that its impact on fatigue performance is relatively small. S_{max} , ΔS ($\Delta S = S_{max} - S_{min}$), and S_m ($S_m = (S_{max} + S_{min})/2$) all show a negative correlation, with S_{max} having the impact on fatigue life, suggesting that concrete is more prone to fatigue failure under high stress conditions. Compared to S_{max} , ΔS has a more pronounced effect on concrete fatigue life. However, the correlation analysis is only a linear dependence analysis, which does not fully consider all the interactions between features and the effects of nonlinearity [53].

It should be noted that the current database consists exclusively of constant amplitude fatigue test results. While this approach is consistent with most design codes and allows for direct comparison with established S-N curves, actual engineering loads—such as wind and wave actions on wind turbine towers—are inherently variable in amplitude. Under variable amplitude loading, the sequence and magnitude of stress cycles can significantly affect fatigue life, often accelerating damage due to high-stress excursions. The equivalent relationship between constant and variable amplitude fatigue results is typically established through damage accumulation rules, such as the Palmgren-Miner linear damage hypothesis. However, these rules have limitations when applied to concrete due to its nonlinear damage evolution and stress-level-dependent behavior.

2.2. The building of backpropagation neural networks model

A backpropagation neural network (BPNN) is a type of multi-layer feedforward network trained using the backpropagation algorithm. [54]. The BPNN consists of an input layer receiving external data, hidden layers extracting features through nonlinear transformations, and an output layer generating final predictions. Connections between neurons, represented by weights, and biases adjust activation thresholds. Activation functions introduce nonlinearity. A loss function measures prediction accuracy, optimized through algorithms like gradient descent. These components form the core structure, adaptable to various problems and tasks in different types of neural networks. The neural network structure used in this paper, as shown in Fig. 3, consists of an input layer, three hidden layers, and an output layer. The input layer has three nodes corresponding to the average compressive strength of concrete, the upper limit, and the lower limit of the stress ratio. The output layer has one node, corresponding to the fatigue life of concrete.

Due to the significant impact that network scale has on improving model generalization and accelerating convergence, a preliminary analysis revealed that hidden layer 1 and hidden layer 3 exhibit better robustness when they are relatively smaller compared to hidden layer 2. Therefore, this study studied the training effects of 21 different scales of hidden layers, with the main parameters as follows: (1) The number of neurons in hidden layer 1 increases from 10 to 410 in increments of 20. (2) The number of neurons in hidden layer 2 increases from 10 to 1010 in increments of 50. (3) The number of neurons in hidden layer 3 increases from 10 to 410 in increments of 20. This resulted in a total of 21 different network models. The model grid parameters are listed in Table 3.

The dataset was divided into a training set and a testing set at a ratio of 9:1. The training set was split into 141 batches and each containing 128 data points, while the testing set was divided into 32 batches and each containing 64 data points. The number of iterations per batch is 1000, resulting in a total of 141,000 iterations. The initial learning rate of the BPNN was set to 0.001, and it is adjusted by a factor of 0.5 every time the iteration count reaches 25% of the total iteration. The model employs the Adam optimizer to minimize the loss function, with Tanh as the activation function. Based on the above settings, the MSE (Mean Squared Error) of the model on the testing set is shown in Fig. 4. According to the test results, this study selected the 13th network structure for this study.

In this study, the implementation of BPNN is carried out in Python 3.11 with the torch library. The test platform includes a Dell desktop with 11th Gen Intel(R) Core (TM) i7-11700@3.60 GHz and 64 G RAM memory.

Table 2
Overview of original and expanded data.

	Raw concrete data				Expanded concrete data			
	$f_{ck}(\text{MPa})$	S_{min}	S_{max}	$\text{Log}_{10}(N)$	$f_{ck}(\text{MPa})$	S_{min}	S_{max}	$\text{Log}_{10}(N)$
Average	98.6	0.166	0.777	4.195	98.879	0.165	0.777	4.196
Max	226.0	0.836	0.960	7.805	226.578	0.838	0.967	7.805
Min	24.0	0.000	0.435	0.477	23.500	0.000	0.432	0.477

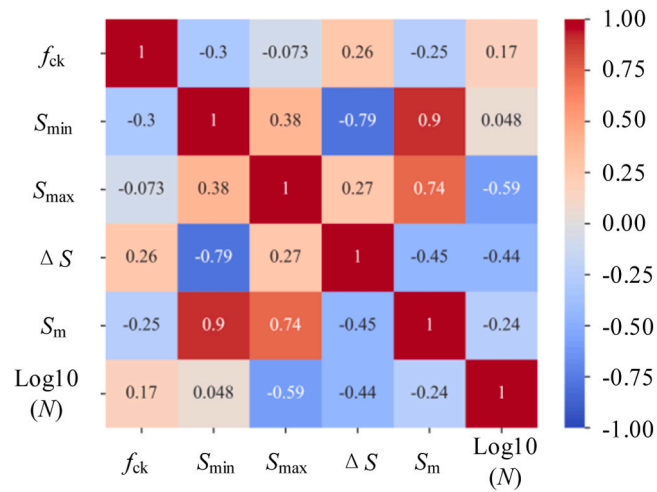


Fig. 2. The heat map of correlation coefficient.

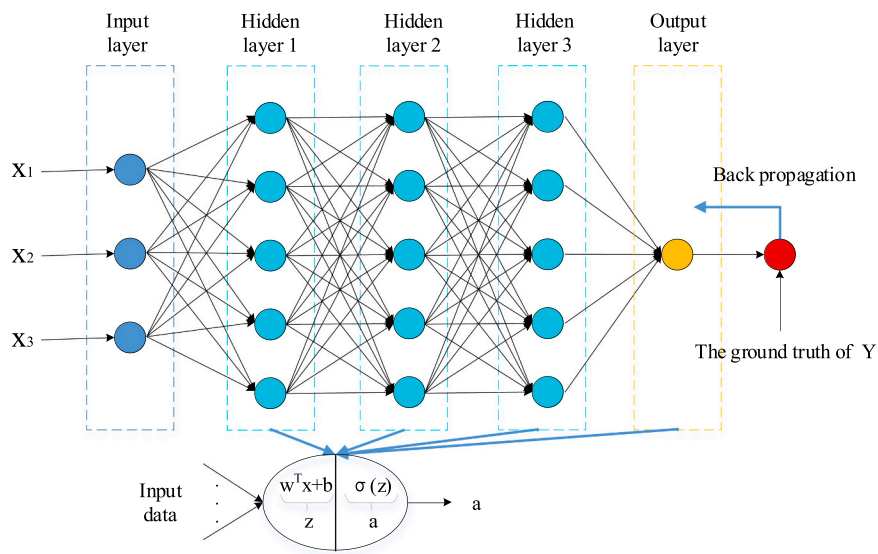


Fig. 3. The structure of the Backpropagation Neural Network.

Table 3

The model grid parameters of BPNN.

Index	Hidden layer 1	Hidden layer 2	Hidden layer 3	Index	Hidden layer 1	Hidden layer 2	Hidden layer 3
1	10	10	10	11	210	510	210
2	30	60	30	12	230	560	230
3	50	110	50	13	250	610	250
4	70	160	70	14	270	660	270
5	90	210	90	15	290	710	290
6	110	260	110	16	310	760	310
7	130	310	130	17	330	810	330
8	150	360	150	18	350	860	350
9	170	410	170	19	370	910	370
10	190	460	190	20	390	960	390
				21	410	1010	410

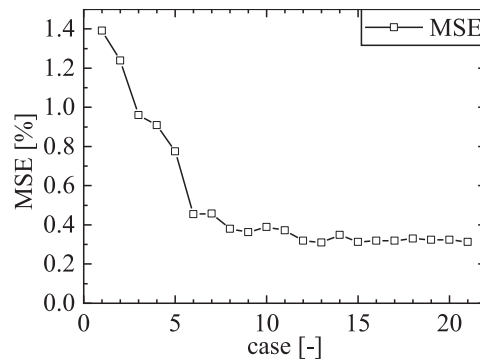


Fig. 4. Mean Squared Error (MSE) of networks with different scales.

2.3. Experimental settings

The concrete tower structure investigated in this study is based on a certified wind turbine tower, with its design methodology referencing the IEC standard series [55,56] and established design practices [57]. To verify the calculation effects of different concrete fatigue S-N curves, a 1:5 scale specimen of a concrete tower was designed based on the existing tower. The mid-section of the specimen is a circular concrete structure with a height of 500 mm, an outer diameter of 1000 mm, and a wall thickness of 60 mm. Concrete plates with a thickness of 300 mm are placed at the top and bottom of the mid-section. The reinforcement ratio of the mid-section is consistent with actual engineering practices. The experimental design parameters are listed in Table 4. To mitigate scaling effects on the concrete's internal structure—particularly the aggregate-cement interface and fracture process zone—the mix design for the 1:5 scale specimen used a reduced maximum aggregate size while maintaining the same water-cement ratio and strength grade as the prototype. This approach ensures that the relative stress-strain behavior and failure mechanisms remain comparable to those of the full-scale structure.

At the top of the specimen, two bundles of prestressed steel strands, each containing five strands, are used to apply prestress. The design stress of the steel strands is 792.87 MPa, resulting in a total prestress of 1110 kN. The steel strands also have a 170 mm eccentricity in the horizontal direction to simulate the eccentric compression state of the tower cylinder. Horizontal actuators are eccentrically arranged on the side of the specimen to apply torsional and bending moments, with an eccentricity distance of 150 mm. The fatigue testing system and test specimens are shown in Fig. 6 and Fig. 7.

The fatigue load form is equal-amplitude fatigue loading, and the fatigue loading history is shown in Fig. 8. The load level is increased every 200,000 cycles, and the loading frequency is 11 Hz. It is important to note that the fatigue load calculation is based on the maximum proportion of fatigue load cycles obtained using the Markov Matrix. The stress range (ΔS) and the mean stress (S_m) at the location of maximum compressive stress are maintained at a ratio of 0.12 in every level, consistent with the corresponding proportions of the fatigue loads in engineering.

The T/M and N_p/N_0 of the specimen design are close to those of the actual loads on wind turbine towers. As horizontal force (F) increases, the T/M initially decreases and then tends to stabilize (Fig. 9) at a predetermined value determined based on the actual loads on the tower. The N_p/N_0 , L_T and M_{top}/N_p of the specimen are determined based on the tower loads and the prestress.

2.4. Finite element model (FEM) for fatigue calculation

This study employs fe-safe for fatigue finite element verification. The results that can be obtained from a Finite Element Model

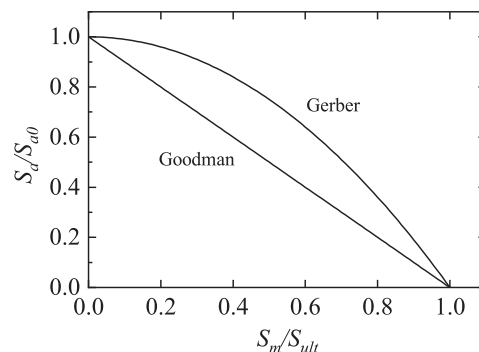


Fig. 5. Comparison of Goodman and Gerber methods.

Table 4
Designs of the tested specimen.

Parameter	Value	Description	Meter
D_{out}	1000	the outer diameter of the specimen's cylindrical body	[mm]
D_{inner}	880	the inner diameter of the specimen's cylindrical body	[mm]
t	60	the thickness of the specimen's cylindrical body	[mm]
h	500	the height of the specimen's cylindrical body	[mm]
N_p/N_0	0.22	the axial compression ratio. N_0 is the cross-sectional compressive strength.	[-]
M_{top}/N_p	170	the bending-compression ratio of the specimen's top surface	[mm]
T/M	0.23	the extreme torsional-bending ratio of the specimen's bottom surface	[-]
L_T	150	the torsion arm of the hydraulic actuator	[mm]
p_s	0.511	the longitudinal reinforcement ratio	[%]
p_{sv}	0.471	the hoop reinforcement ratio	[%]
$f_{c,mean}$	78.6	the mean value of the concrete compressive strength determined through testing	[MPa]
f_y	360	the design value of the tensile strength of the rebar	[MPa]

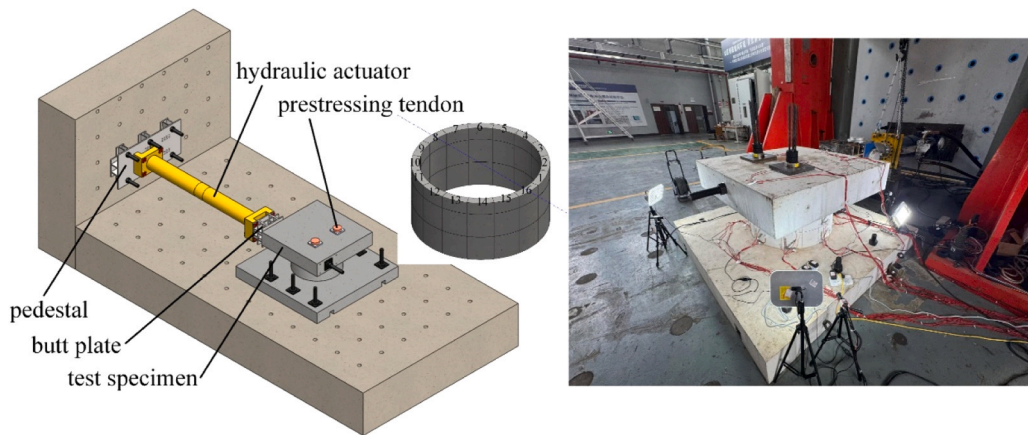


Fig. 6. The fatigue testing system.

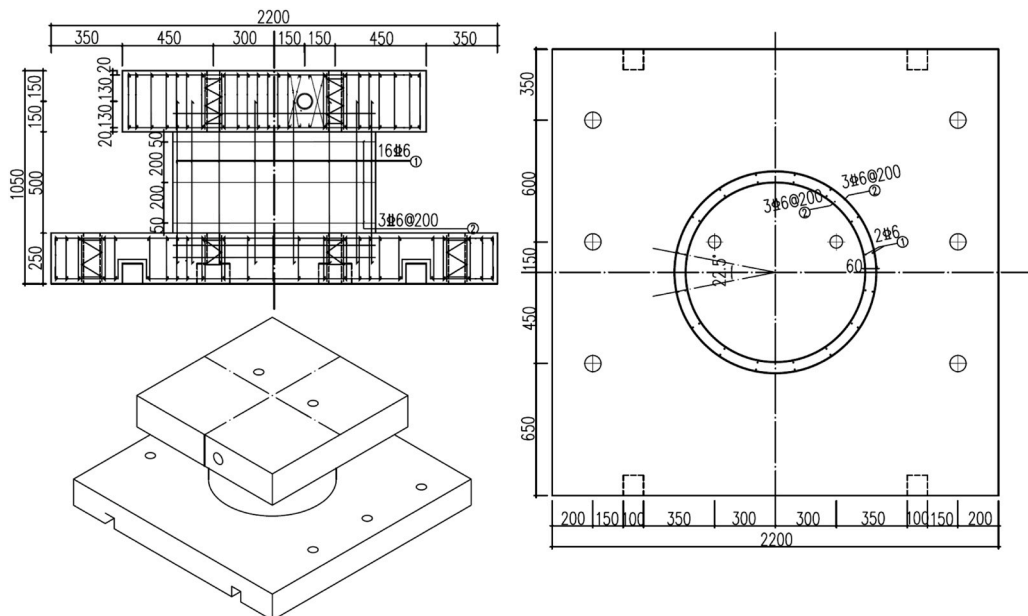


Fig. 7. The size of the test specimen.

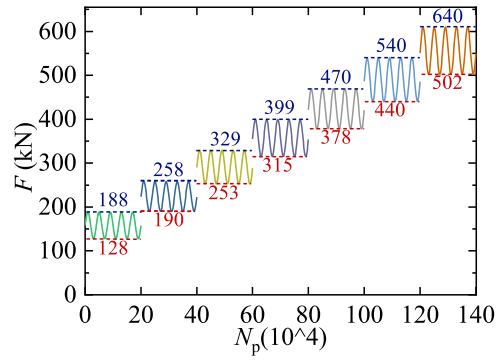


Fig. 8. The fatigue loading history.

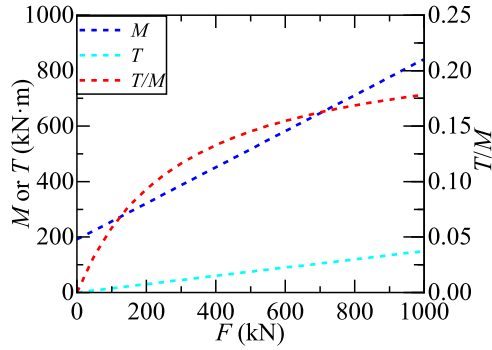


Fig. 9. The relationship between M and T .

include stress, strain, temperature, deflection, and frequency response, which can be used for fatigue analysis. Several fatigue assessments rely on stress data derived from linear elastic Finite Element Analysis (FEA). This type of analysis operates under the assumption of a linear correlation between load and response. The nonlinear behaviors arising from material and geometric effects are obtained through interpolation of load step results derived from finite element analysis (FEA) computations. Concrete materials, as typical nonlinear and anisotropic materials, fatigue performance not only depends on the stress amplitude but also on the average stress and the direction of stress. In this study, three methods for mean stress corrections are considered: Goodman, Gerber [58] and R ratio S-N curves [59] (Multiple S-N curves dependent on stress ratio). The equivalent stress amplitude (S_a) can be converted using the Goodman (Eq.13) and Gerber relations (Eq.14), and the Goodman and Gerber curves are compared in Fig. 5.

$$\frac{S_a}{S_{a0}} + \frac{S_m}{S_{ult}} = 1 \quad (13)$$

$$\frac{S_a}{S_{a0}} + \left(\frac{S_m}{S_{ult}} \right)^2 = 1 \quad (14)$$

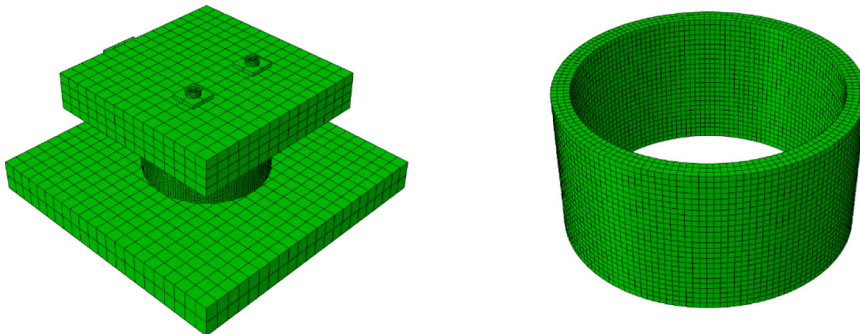
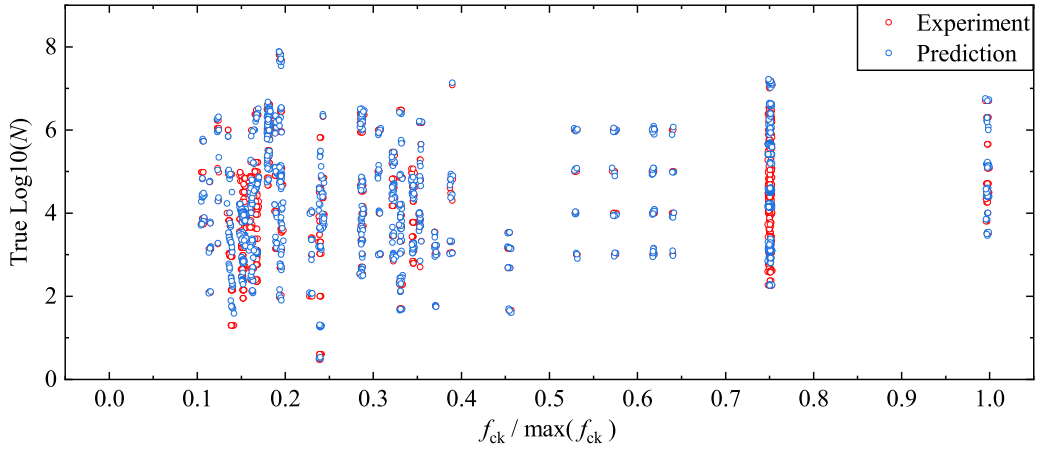
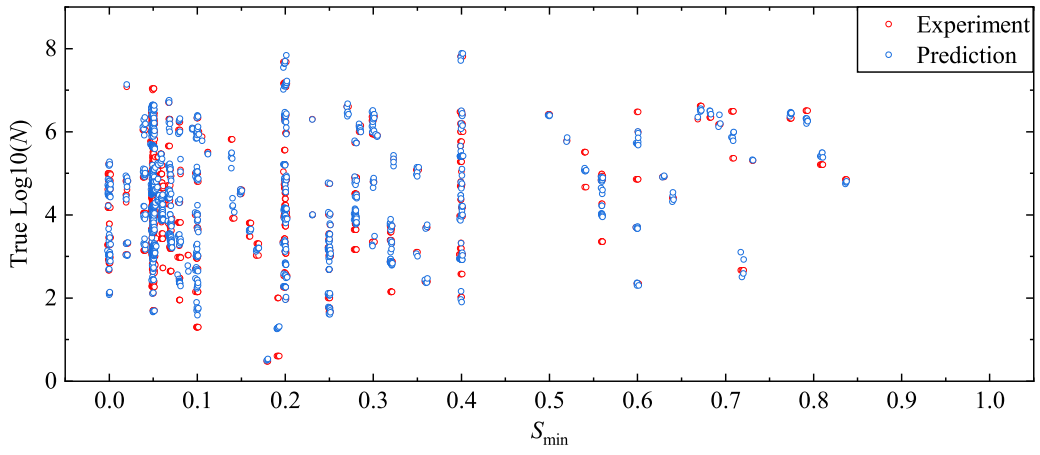


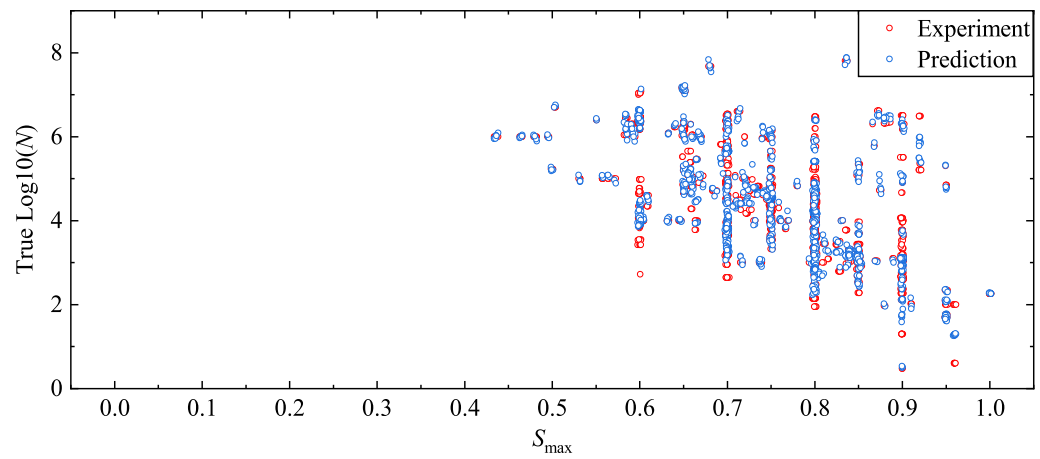
Fig. 10. The finite element model of the specimen.



(a)



(b)



(c)

Fig. 11. The verification of BPNN. (a) Concrete Strength, (b) Minimum stress ratio, and (c) Maximum stress ratio.

In this equation, S_a is the equivalent stress amplitude, S_{a0} is the stress amplitude at zero mean stress, S_m is the mean stress and S_{ult} is the material ultimate strength.

Concrete tensile fatigue is relatively low, this study considers concrete fatigue failure when concrete reaches its tensile strength. So, this study only considers compressive fatigue and avoids using stress beyond the tensile limit in fatigue calculations by employing a concrete tensile damage plasticity constitutive model in FEA. By explicitly incorporating concrete tensile failure mechanisms into the FEA, fe-safe simulations can effectively characterize fatigue behavior under design-level stress states through stress interpolation across load steps. FEM of the specimen is shown in Fig. 10. The S-N curves used in the finite element analysis will be obtained respectively from codes and BPNN in this study, to verify the effectiveness of these methods in analyzing experimental data.

FE-safe is an advanced fatigue durability analysis and signal processing software. As a fatigue life analysis package based on finite element models, it stands as the industry leader in multiaxial fatigue analysis solutions, renowned for its sophisticated algorithms, comprehensive functionalities, and unparalleled precision-globally acknowledged as the most accurate fatigue analysis software. The platform uniquely supports dual approaches: fatigue analysis techniques grounded in stress/strain signals from experimental testing, and fatigue simulation design technologies powered by finite element analysis (FEA) computations. This dual capability bridges experimental validation with computational modeling, establishing FE-safe as an indispensable tool for both research and industrial applications in structural durability assessment. The fatigue analysis results in this study were derived from FE-safe.

It is important to note that the BPNN model does not include prestress as an input parameter, as the training data were derived from published experimental studies that did not systematically vary prestress levels. However, in the finite element analysis conducted in fe-safe, the prestress force was explicitly applied to the model to accurately represent the stress state in the concrete. This allows the fatigue analysis to account for the mean stress elevation caused by prestressing, which is critical for realistic life prediction. The mean stress correction methods (Goodman, Gerber, and R-ratio) further incorporate this effect by adjusting the equivalent stress amplitude based on the local mean stress.

3. Results and discussion

3.1. Comparison between BPNN and codes

The results of BPNN on the training set is shown in Fig. 11. A max-normalization procedure was applied to f_{ck} to enhance the interpretability of prediction result distributions. From results, it can be observed that most of the predicted data fit well with the experimental data, and the predictions at discrete points also align with the experimental results. Considering the concrete strength (Fig. 11(a)), it can be observed that some prediction results within the range of 0.15–0.16 are more conservative compared to the

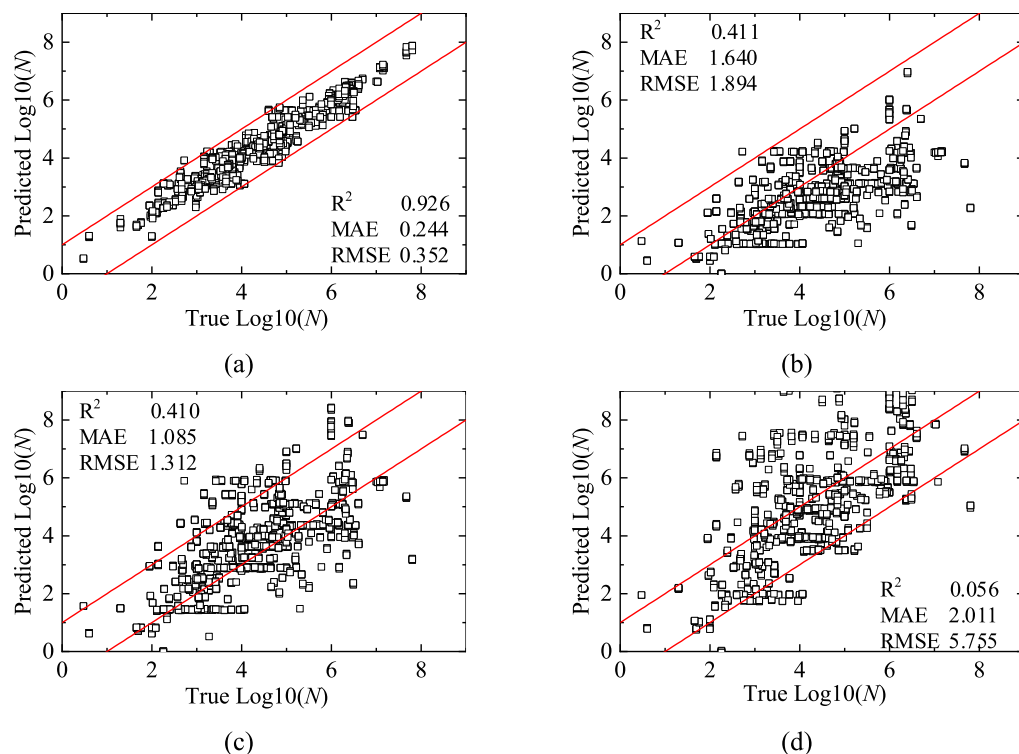


Fig. 12. The comparison of predicted outcomes: (a) BPNN, (b) NEN 6723, (c) EN 1992, and (d) fib.

experimental results, as indicated by the $\text{Log}_{10}(N)$ values clustering at a lower level. From the perspective of $S_{\min}$ and $S_{\max}$, most of the predicted data points align well with the experimental results.

To evaluate the prediction performance of these methods for concrete fatigue life, this paper compares BPNN with three existing methods proposed by NEN 6723, EN1992, and fib (see Table 1). The results of the four models on the test datasets are shown in Fig. 12. We can observe that the computational model in NEN 6723 is conservative, because most data points indicate that the predicted life is shorter than the actual life. In contrast, EN 1992 and fib are more aggressive compared to NEN 6723 but provide a correct reflection of the concrete's fatigue life in most cases. Among these models, the BPNN performs best, showing no significant order-of-magnitude discrepancies and aligning more closely with experimental results. This is due to the powerful capability of machine learning algorithms to process large datasets, consider more influencing factors and uncover hidden relationships within the data.

To further analyse the relationship between concrete fatigue life, concrete compressive strength, and stress, the three-dimensional graph is plotted as shown in Fig. 13. We can observe that the concrete fatigue life increases when f_{ck} increases and ΔS decreases across all prediction methods. The prediction results obtained from the BPNN are better distributed, effectively excluding the influence of invalid data.

Based on the analysis results, we can conclude that BPNN demonstrates better performance in predicting concrete fatigue life and does not overestimate or underestimate it excessively. Overestimation would lead to overly conservative calculations that cannot fully utilize the material properties, while underestimation would result in overly aggressive calculations that may cause structural damage. Additionally, the fitting results from the BPNN are better robustness. This makes BPNN suitable for determining the S-N curve of concrete, enabling more complex analyses.

3.2. Result of experiment

According to the loading regime described in Section 2.3, the specimen failed after 1.24 million load cycles. The limited crack width development in test specimens posed significant observational challenges. To address this, crack propagation was traced using post-processing techniques applied to high-resolution photographic data. Fig. 14 documents newly observed crack patterns corresponding to completion points of specific loading cycles, with geometric features quantified using pixel-scale resolution analysis. Cracks first appeared in Zone 14 at 0.4 million cycles, and these cracks had expanded by 0.6 million cycles. When the load reached 1 million cycles, multiple diagonal cracks, approximately 10 cm in length and widely dispersed, began to appear in Zones 11–13. By 1.2 million cycles, the cracks had gradually coalesced and the opening and closing of cracks were evident during the loading process. After the start of the

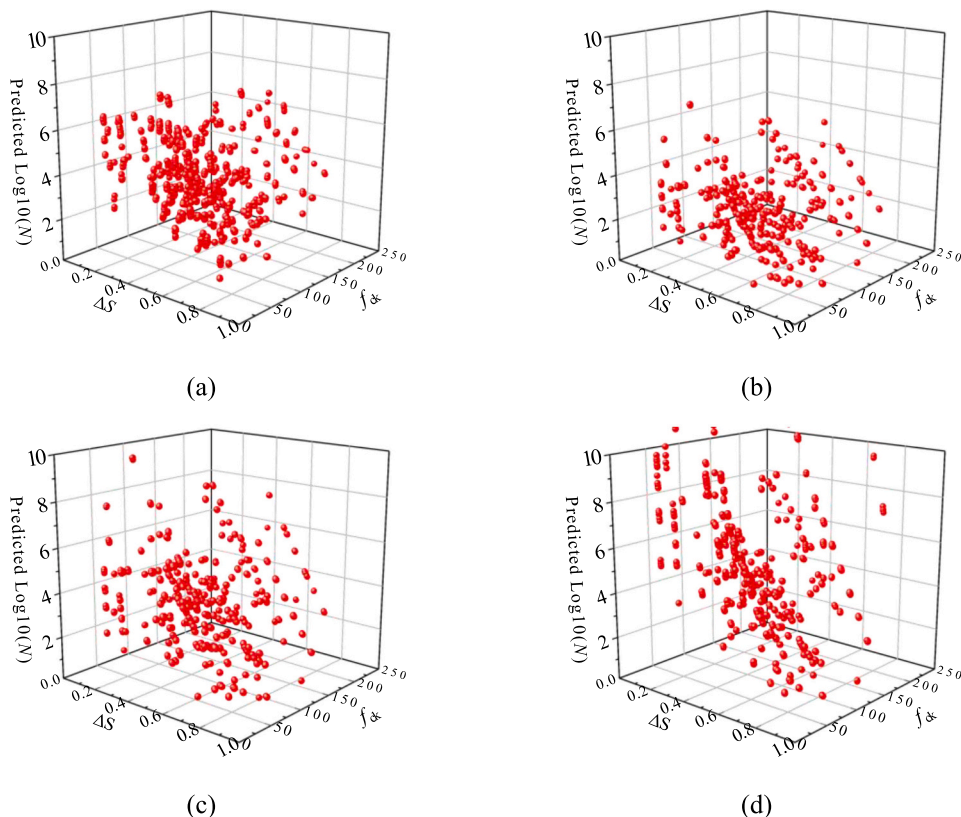


Fig. 13. Three-dimensional graph of concrete fatigue life. (a) BPNN, (b) NEN 6723, (c) EN 1992, and (d) fib.

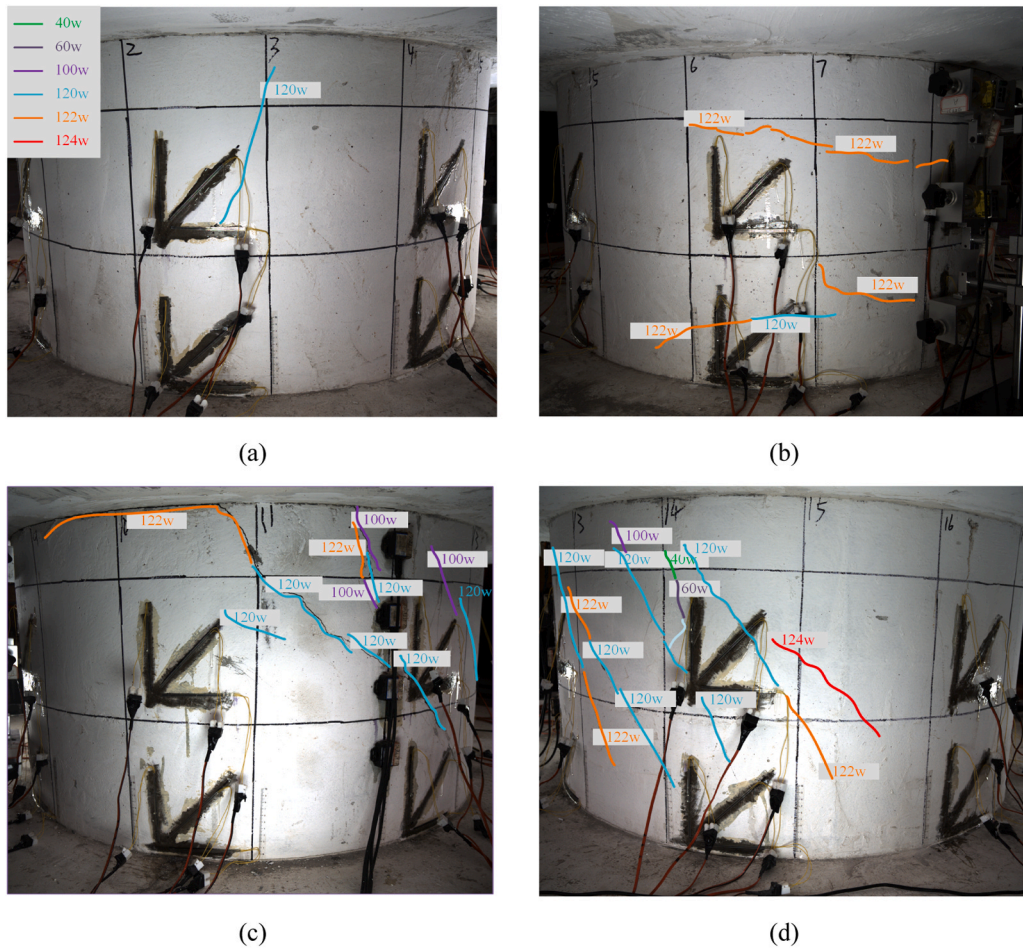


Fig. 14. The crack development of the specimen.

1.2 millionth cycle, horizontal cracks appeared in Zones 6–7, and the diagonal cracks in Zone 11 widened. After a period of stabilization, these cracks rapidly led to failure at 1.24 million cycles. From Fig. 15, it can be observed that the concrete at the bottom of Zones 13–3 was crushed at the time of failure, and there were dense diagonal cracks in Zones 11 to –14. The top of Zones 4–7 experienced failure, accompanied by localized concrete spalling.

Fatigue failure in this study is defined as the complete loss of structural load-carrying capacity under sustained cyclic loading, which physically corresponds to a reduction in bearing capacity exceeding 15%. This threshold aligns with established definitions in concrete fatigue design codes, where a 15% degradation in stiffness or strength is typically regarded as the serviceability limit and the onset of structural failure (fib Model Code 2010). However, due to the inherently brittle nature of high-strength concrete under compressive fatigue, the specimen exhibited sudden and irreversible failure upon reaching 1.24 million cycles, characterized by immediate crushing and spalling without a prolonged period of gradual degradation. This behavior precluded the incremental capture of the exact 15% degradation point using conventional sensors during loading. Post-test observations confirmed that the failure state corresponded to degradation well beyond this threshold.

3.3. Application in *fe-safe*

In *fe-safe*, the Goodman and Gerber methods use S-N curves with a mean stress of zero and convert them into equivalent stress amplitude calculations using the corresponding formulas. The R ratio S-N curves are computed for different R values, and interpolation is performed in *fe-safe* to cover a range of R values. The methods provided by the codes can easily calculate the required S-N curves, but BPNN does not provide an explicit functional relationship for concrete fatigue life calculations. A trained network can compute point mappings for a given concrete material, which can also be applied in finite element analysis. The trained Backpropagation Neural Network (BPNN) from Section 2.2 was employed to develop S-N curves, with subsequent implementation into material configuration parameters of *fe-safe* to enable fatigue life prediction.

When using the Goodman and Gerber methods, the S-N curve shown in Fig. 16(a). For the R ratio S-N curves method, the S-N curves shown in Fig. 16(b-d), and the S-N curves obtained by BPNN are compared with those from the code methods. In Fig. 16(a), EN 1992

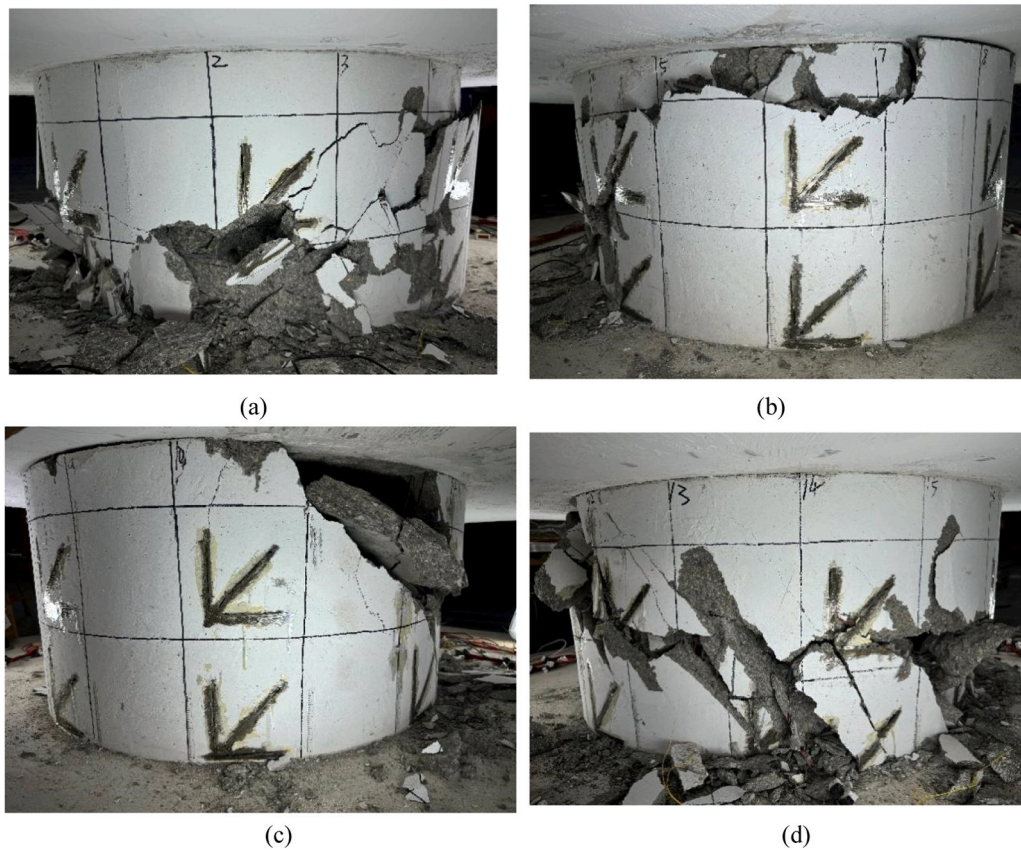


Fig. 15. The failure of the specimen.

provides the most conservative S-N curve among the standard methods, whereas BPNN is more aggressive in the low-cycle regime but more conservative in the high-cycle regime. For different R values in the S-N curves, the stress amplitude S_a decreases as R increases for the same number of cycles. The BPNN method also shows higher load-bearing capacity in low-cycle fatigue and lower load-bearing capacity in high-cycle fatigue compared to the code methods.

Notably, the S-N curves for different R values intersect at lower $\log N$ values for the BPNN method, indicating that higher mean stress levels improve material fatigue performance in high-cycle fatigue, which is why the standards consider this more conservatively. Additionally, when $\log N$ is greater than 5, the BPNN method becomes more conservative compared to the standard methods. In the high-cycle fatigue regime ($\log N > 5$), the BPNN curves show that higher mean stress (higher R) leads to improved fatigue performance, which is consistent with the physical understanding that compressive mean stress can inhibit crack opening and propagation in concrete. However, the intersection at low cycles suggests that in the low-cycle fatigue regime, the beneficial effect of mean stress is diminished or even reversed. From a physical perspective, under high stress amplitudes (low-cycle fatigue), microcracking and damage accumulate rapidly, and the mean stress level may have a less pronounced effect compared to the stress amplitude.

This study uses the S-N curves calculated according to the fib to compute the fatigue analysis for three different mean stress treatment methods to determine the most suitable mean stress treatment approach for concrete materials. In fe-safe, all cyclic loads at the time of experimental failure are used to calculate the specimen life (LOGLife) and the factor of strength (FOS). LOGLife refers to the logarithm of the number of repeatable cycles of all cyclic loads. The factor of strength (FOS) is the factor that, when applied to either the loading or to the elastic stresses in the finite element model, will produce the required target life at the node.

As shown in Fig. 17, the fatigue life calculation results using the Goodman and Gerber methods indicate that the tower experiences fatigue life exceedance at localized positions, and the FOS calculation results are similar, which is clearly insufficient to cause comprehensive structural failure. The R ratio calculation results show that more than two-thirds of the tower exceeds the fatigue life limit, which is closer to the experimental results. The FOS calculation results also show that, apart from the already damaged positions, some units are in a state of imminent failure. Therefore, the R ratio method is more appropriate for handling the effect of mean stress on the fatigue life of concrete materials, while the Goodman and Gerber methods do not adequately account for mean stress.

Therefore, this study uses the R ratio method to handle the effect of mean stress on the fatigue life of concrete materials. In fe-safe, the LOGLife and FOS of the specimens are calculated using four different S-N curves. The calculation results for the S-N curve defined in fib are shown in Fig. 17(c), and the others are shown in Fig. 18. According to the calculation results of the R ratio method, the methods are ranked from the most severe to the least severe damage as BPNN, NEN6729, EN1992, and fib, indicating that the BPNN method is

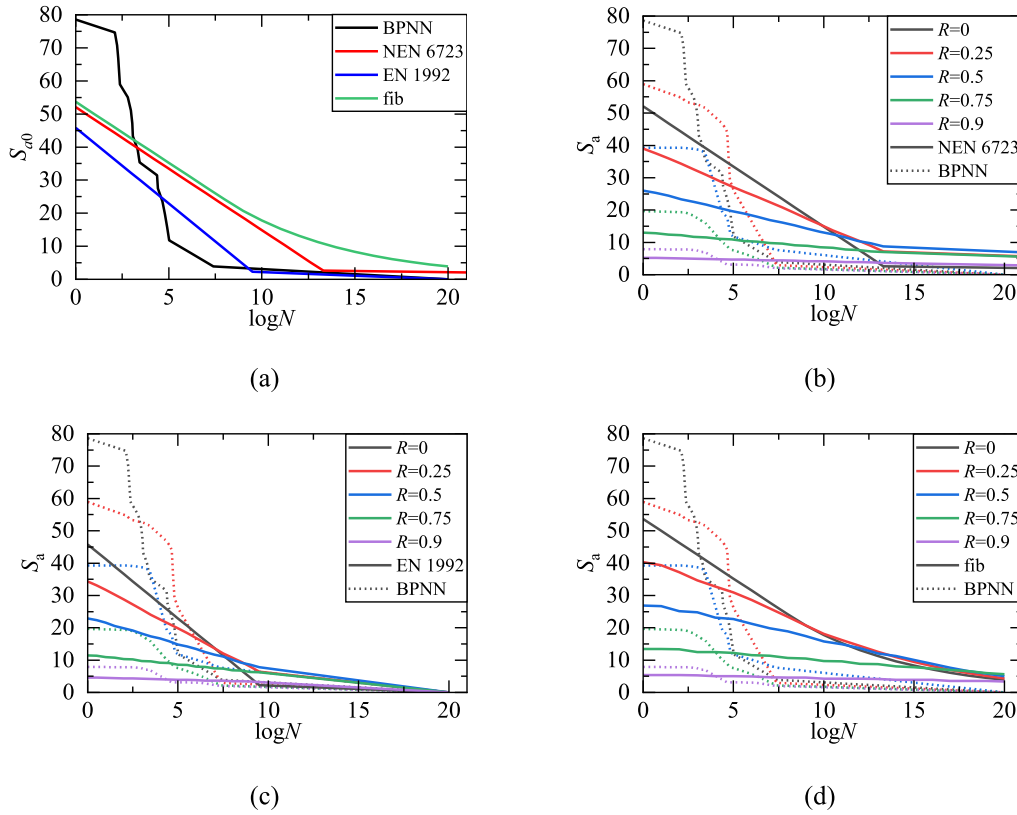


Fig. 16. The S-N curve used in fe-safe. (a) The S-N curve at zero mean stress, (b) NEN 6723, (c) EN 1992, and (d) fib.

the most conservative. Compared to the compressive failure regions of the specimens, the BPNN method is more sensitive to stress variations and shows nearly complete fatigue failure in all units, while the fatigue failure regions calculated by the other standards are similar.

3.4. Key curves

The load-displacement curves and stiffness degradation curves obtained from fatigue finite element analysis and experimental testing are shown in Figs. 19 and 20, respectively. The average error in structural stiffness between the finite element model and the experimental results is 9.43%. From Figs. 19 and 20, it can be observed that during the initial 0–200,000 fatigue loading cycles, the load-displacement curve obtained from the experiment exhibited significant slip, and the stiffness showed an initial decrease followed by an increase. In contrast, the structural stiffness obtained from the finite element analysis demonstrated a steadily decreasing trend. This discrepancy is attributed to false displacement occurring during the initial stage of experimental fatigue loading.

At each load level increase (every 200,000 cycles), the structural stiffness initially increased during preloading (as the structure entered a new equilibrium state), then gradually decreased during fatigue loading before eventually stabilizing. This behavior indicates progressive damage accumulation in the structure throughout the fatigue process. From 1.0 million cycles to failure, the average stiffness degradation rate was 3.25 kN/(mm·10⁵ cycles), compared to only 0.452 kN/(mm·10⁵ cycles) during the 0.2–0.8 million cycle range. These observations demonstrate that the development of fatigue damage in the structure follows three distinct stages: stress redistribution, slow accumulation of fatigue damage, and rapid failure.

4. Summary and conclusions

This study uses an effective BPNN for predicting the S-N curve of concrete, and BPNN was developed and trained using 558 experimental results. The experimental validation was conducted based on the geometry and loads of wind turbine towers, and calculations were performed using four different S-N curves from BPNN, NEN 6723, EN 1992, and fib, with appropriate mean stress treatment methods chosen in fe-safe. Based on this work, the following conclusions can be drawn:

- (1) Compared to existing standard methods, the BPNN achieves superior accuracy ($R^2 = 0.926$, $MAE = 0.244$, $RMSE = 0.352$) in fitting the fatigue life of concrete from an expanded experimental database. The BPNN effectively extracts hidden nonlinear

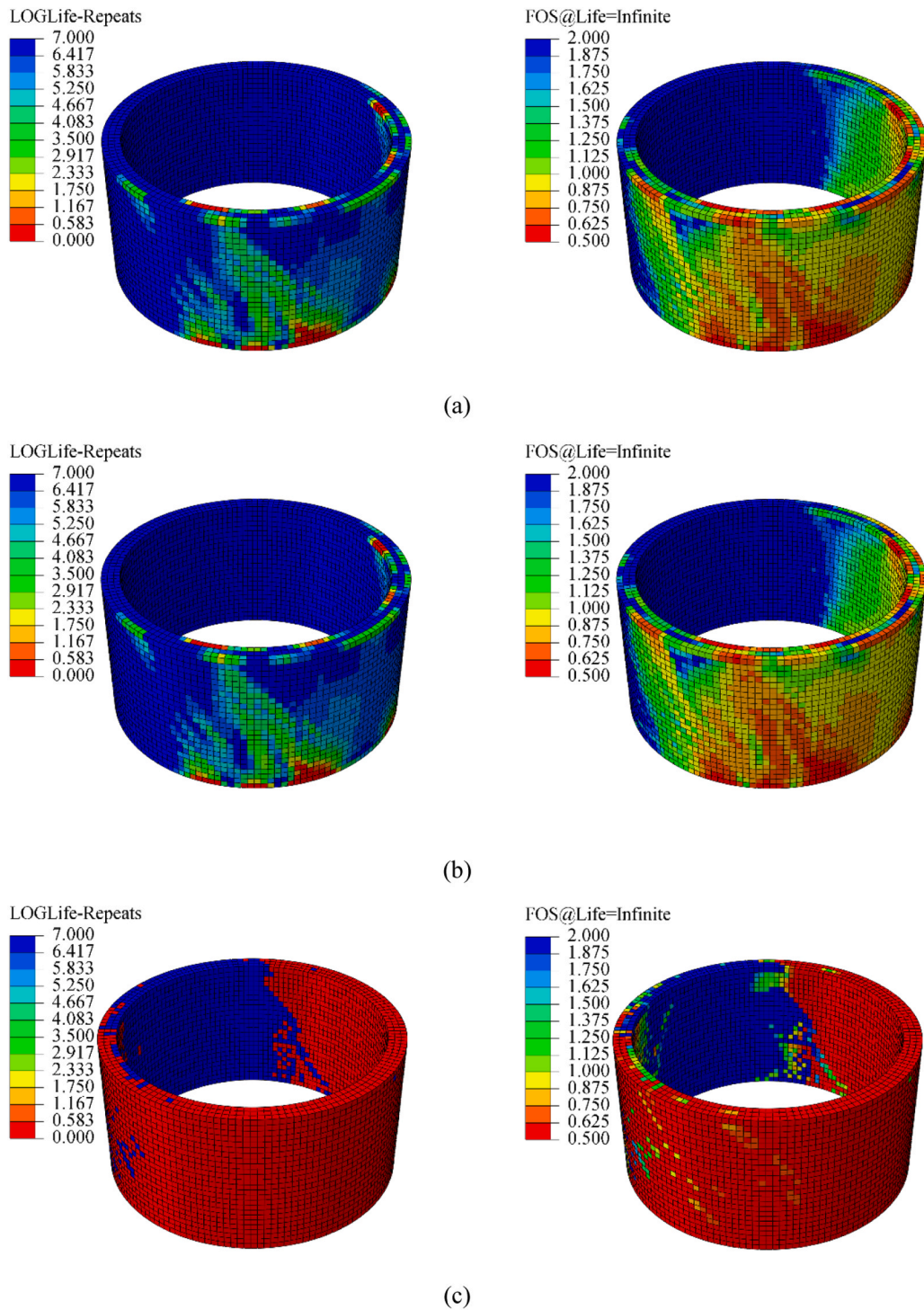
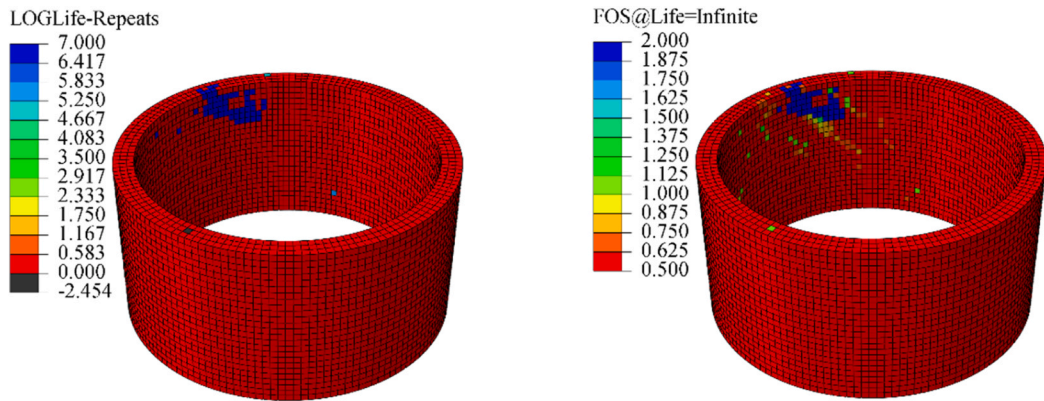


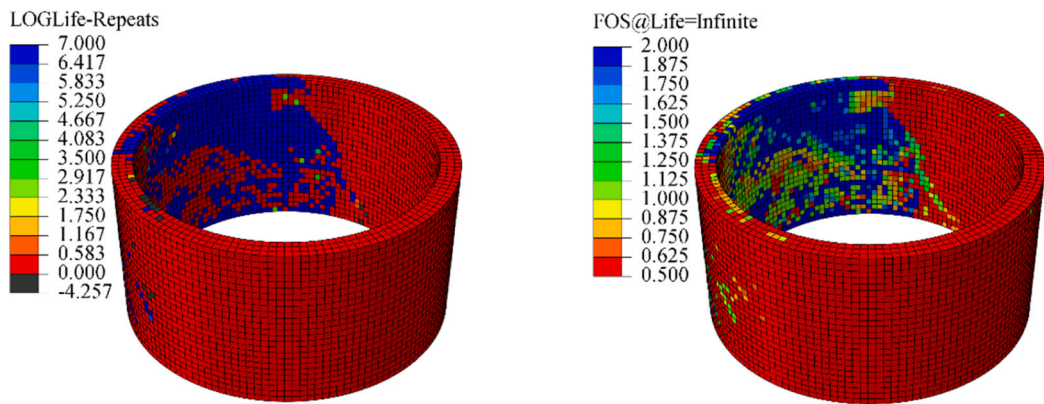
Fig. 17. LOGLife and FOS of different methods. (a) Goodman, (b) Gerber, and (c) R ratio.

relationships between concrete strength, stress levels, and fatigue life that are not captured by conventional code-based S-N curves.

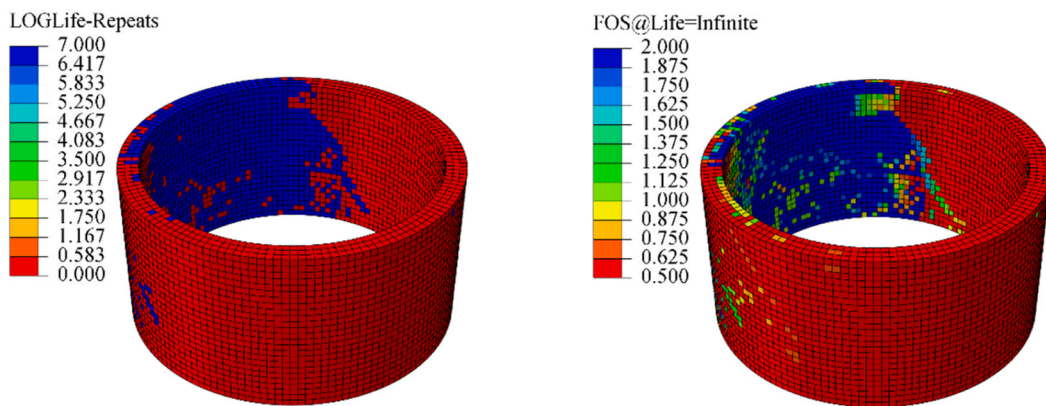
- (2) Through systematic evaluation in fe-safe, the R-ratio method is identified as the most suitable approach for handling mean stress effects in concrete fatigue. The Goodman and Gerber methods significantly overestimate fatigue performance and are unsuitable for concrete.



(a)



(b)



(c)

Fig. 18. LOGLife and FOS of different S-N curves. (a) BPNN, (b) NEN 6723 and (c) EN 1992.

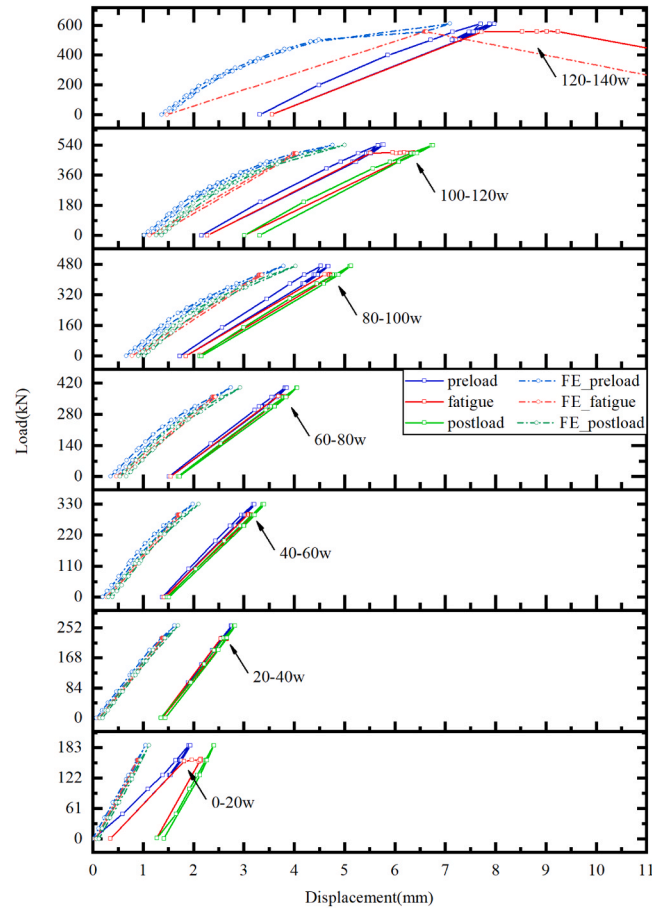


Fig. 19. The load-displacement curves.

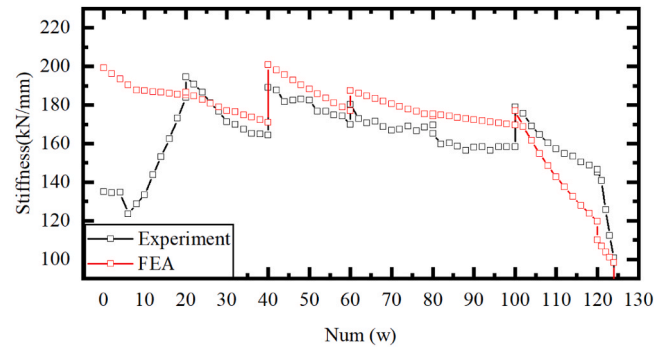


Fig. 20. The stiffness degradation curves.

- (3) When implemented in fe-safe with the R-ratio method, the BPNN-based S-N curve predicts fatigue life within 5% of the experimental value, outperforming all code-based methods. The BPNN is also more sensitive to stress variations and provides conservative estimates in the high-cycle regime, making it suitable for engineering design.
- (4) A limitation of this study is the insufficient amount of high-cycle fatigue test data for concrete materials in the existing literature, which may affect the reliability of BPNN predictions in the very high-cycle regime ($\log N > 7$).
- (5) The fatigue damage evolution of the concrete tower structure exhibits a distinct three-stage process: initial stress redistribution with stiffness fluctuation, steady damage accumulation with gradual stiffness degradation, and rapid failure with accelerated stiffness loss. The finite element model employing the BPNN-based S-N curve with R-ratio means stress correction successfully captures this overall degradation trend, achieving an average stiffness error of 9.43%.

This contribution indicates that BPNN can effectively and accurately predict the fatigue life of compressed concrete using experimental datasets collected from the literature, and the algorithm can be supplemented and updated with additional test data based on the original database to enhance the applicability and performance of concrete fatigue life prediction.

Future work will focus on extending the BPNN framework to incorporate variable amplitude loading effects by training on datasets that include sequence effects and by integrating cumulative damage models. This will enable more accurate predictions of concrete fatigue life under realistic loading conditions.

Nomenclature

D_{inner}	the inner diameter of the specimen's cylindrical body
D_{out}	the outer diameter of the specimen's cylindrical body
$E_{\text{cd, max}}$	the stress level
F	the horizontal force of the hydraulic actuator
$f_{\text{b,v}}$	the fatigue reference strength of concrete
$f_{\text{b,rep,k}}$	the characteristic value of the uniaxial short term concrete compressive strength ($=0.85f_{\text{ck}}$)
$f_{\text{b,rep,v}}$	the characteristic value of the concrete compressive strength in the limit state of fatigue
$f_{\text{c,mean}}$	the mean value of the concrete compressive strength determined through testing
$f_{\text{cd,fat}}$	design fatigue strength
f_{ck}	characteristic compressive cylinder strength of concrete at 28 days
f_y	the design value of the tensile strength of the rebar
h	the height of the specimen's cylindrical body
L_T	twisting force arm
M	the bending moment of the test specimen's bottom
M_{top}	the bending moment at the top of the test specimen caused by prestress eccentricity
N	number of cycles to failure
N_p	the pressure caused by prestress.
N_0	the design value of the test specimen's compressive bearing capacity
P	probability of failure
R	the stress ratio
S_a	the equivalent stress amplitude
S_{a0}	the stress amplitude at zero mean stress
S_m	the mean value between S_{min} and S_{max}
$S_{\text{min}}, S_{\text{max}}$	the maximum and minimum fractions of compressive strength applied in load cycles
S_{ult}	the material ultimate strength
s	a factor that depends on the strength class of the cement
T	the torque of the test specimen
t	the thickness of the specimen's cylindrical body
t_0	the time of the start of the cyclic loading on the concrete
$\beta_{\text{cc}}(t_0)$	coefficient for the concrete strength at first load application
γ_m	the partial factor of the material
ΔS	the difference between S_{min} and S_{max}
ρ_s	the longitudinal reinforcement ratio
ρ_{sy}	the hoop reinforcement ratio
$\sigma_{\text{max}}, \sigma_{\text{min}}$	design value of the maximum and minimum compressive stress in the concrete

Abbreviation

ANN	artificial neural networks;
BPNN	backpropagation neural network;
BR-BPNN	Bayesian regularized backpropagation neural network;
FEA	finite element analysis;
FEM	finite element model;
FOS	factor of strength;
MSE	mean squared error;
MAE	mean absolute error;
S-N	stress-life curve;
UHPC	ultra-high performance concrete.

CRedit authorship contribution statement

Tianhao Li: Data curation. **Lin Yang:** Writing – review & editing, Visualization, Methodology, Conceptualization. **Jiawei Zhang:** Data curation. **Kun Fu:** Writing – review & editing. **Yuhang Wang:** Funding acquisition. **Dongliang Zhang:** Data curation. **Huang Cirong Huang:** Writing – original draft, Data curation. **Junlin Li:** Data curation.

Declaration of Competing Interest

The authors declare no conflicts of interest.

Data availability

Data will be made available on request.

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